

Brennan Duck

LinkedIn | GitHub | Portfolio | Knoxville, TN

Education

Georgia Institute of Technology | *M.S. Computer Science* | **Incoming Fall 2026**

- Specialization: Computing Systems (Planned)

Western Governors University | *B.S. Computer Science* | **Expected May 2026**

- Focus: Data Structures, Algorithms, Operating Systems, Computer Architecture

Technical Skills

- **Languages:** Go (Golang), C, C++, Python, SQL, Bash, Java, TypeScript
- **Backend & Systems:** RESTful APIs, Distributed Systems, Concurrency, Memory Management, TCP/IP, Relational Databases
- **Cloud & Infrastructure:** AWS (S3, Lambda, EC2, IAM), Docker, Coolify, Linux/Unix, Git, PostgreSQL

Projects

Redish - Distributed Key-Value Store | *Go, TCP, Concurrency* | **Dec 2025 – Present**

A high-performance, sharded key-value store built from scratch to explore distributed consensus and storage engines.

- Architected a custom TCP wire protocol for client-server communication, bypassing HTTP overhead for lower latency.
- Implemented concurrent read/write operations using mutex locking to ensure thread safety under high load.
- Designed a Write-Ahead Log (WAL) to ensure data durability and crash recovery.

Bonobo Language Interpreter | *Go, AST, Parsing* | **Nov 2025 – Present**

A custom interpreted programming language written in Go.

- Implemented a Pratt Parser (Top-Down Operator Precedence) to handle complex expressions without parser generators.
- Built a REPL and AST-dumper to visualize parsing logic and tokenization in real-time.

Professional Experience

Duck Family Dental | *Digital Systems Lead* | **Jan 2025 – Present**

Architected and deployed internal digital infrastructure for a healthcare provider, focusing on cost optimization and process automation.

- **Cost Optimization:** Migrated operations from costly third-party SaaS to custom internal tools, reducing annual technical overhead by thousands of dollars.
- **Backend Development:** Engineered a custom notification system to integrate internal patient records with external messaging APIs, improving retention and reducing appointment no-shows by ~15%.
- **Infrastructure & DevOps:** Managed a Dockerized Linux environment on self-hosted hardware, handling DNS, SSL rotation, and CI/CD workflows for zero-downtime updates.
- **Compliance:** Ensured strict adherence to HIPAA standards regarding data privacy and security.

Freelance | *Software Developer* | **May 2024 – Present**

- Developed performant web solutions for local businesses, integrating legacy systems with modern RESTful APIs to automate data entry and reporting workflows.

Slopopedia | *Software Engineering Extern* | **March 2024 – May 2024** *A community platform for cult and B-movie enthusiasts*

- Operated in an Agile environment, participating in code reviews and sprint planning to improve development velocity.
- Implemented S3 presigned URL workflows to optimize secure asset delivery for media-heavy pages, reducing load times and eliminating public bucket exposure.